

Mapbox Studio

Setup Guide

Complete reference for configuring tilesets, layers, and overlays in Mapbox Studio for the Rapid Cortex map integration. Covers all verticals: 911 / PSAP, Campus Safety, Venue Operations, and Airport.

LAYER	SOURCE	STATUS
County lines	Mapbox built-in	NO UPLOAD
State boundaries	Mapbox built-in	NO UPLOAD
Airports / Aviation	FAA national — 10,642 features	UPLOAD READY
Agency / PSAP zones	Draw or import	DRAW & UPLOAD
Venue zones	Per client	PER ONBOARDING
Campus zones	Per client	PER ONBOARDING
Live incidents	App GeoJSON at runtime	APP-MANAGED

OVERVIEW

How the Map Integration Works

The Rapid Cortex map is built on two independent layers of data:

STUDIO LAYERS	APP LAYERS
Static geographic context: county lines, state boundaries, agency zones, airports, venue/campus footprints. Configured once in Mapbox Studio and baked into the published style URL.	Live operational data: incident markers, caller/report location pins. Injected at runtime as GeoJSON from the RC API. Always renders on top of Studio layers.

The app code references Studio layers by their exact layer IDs. If a layer ID in Studio does not match map-constants.ts, the toggle control silently skips it — it will not crash, but the layer will not respond to visibility changes. Always verify IDs before publishing.

CANONICAL LAYER IDs

```
# These IDs must match exactly in Mapbox Studio
rc-agency-zones-fill      rc-agency-zones-line
rc-counties-line         rc-state-boundaries-line
rc-airports-circle       rc-airports-label
rc-campus-zones-fill     rc-campus-zones-line
rc-venue-zones-fill      rc-venue-zones-line
rc-live-traffic          rc-live-traffic-closures

# Added by app at runtime — do NOT add these in Studio
rc-live-incidents-circle rc-live-incidents-pulse
rc-live-incidents-resolved rc-caller-location-circle
```

RENAME REQUIRED — Live Traffic Layers

The style currently contains two traffic layers named with display names rather than canonical IDs. Rename them in Studio before publishing so the app toggle controls can reach them:

CURRENT NAME IN STUDIO	RENAME TO (exact)	TOGGLE LABEL IN APP
RC Live Traffic	rc-live-traffic	Live Traffic
RC Live Traffic Closures	rc-live-traffic-closures	Traffic Closures
Rapid Cortex U.S. Counties	rc-counties-line	Counties
RC U.S. State Boundaries	rc-state-boundaries-line	State Boundaries

To rename: click the layer in the Studio panel → click the layer name at the top of the config panel → type the new ID → press Enter.

Traffic layer toggle defaults:

```
liveTraffic:      false ← off by default, dispatcher toggles on when needed
liveTrafficClosures: false ← off by default, toggle on for routing awareness
```

Both traffic layers are available in all four verticals: core (911), campus, venue, and airport. Traffic closures are especially relevant for venue operations on event days and for EMS/fire routing during major incidents.

1 County Lines

No data upload required — uses Mapbox built-in tileset

Open your RC style in Mapbox Studio. Click + to add a new layer. Configure as follows:

```
Type:      Line
Source:     mapbox-streets-v8
Source layer: admin
Filter:     admin_level = 2    (county / parish level in US)
Layer ID:   rc-counties-line
```

Paint properties:

```
line-color:  #2d2a4a
line-width:   0.7
line-opacity: 0.6
```

2 State Boundaries

No data upload required — uses Mapbox built-in tileset

Add another layer from the same mapbox-streets-v8 source, using admin_level = 1 to target state-level boundaries:

```
Type:      Line
Source:     mapbox-streets-v8
Source layer: admin
Filter:     admin_level = 1
Layer ID:   rc-state-boundaries-line
```

Paint properties:

```
line-color:  #3b3760
line-width:   1.2
line-opacity: 0.5
line-dasharray: [4, 2]
```

3 Airports & Aviation Facilities

National dataset — 10,642 operational airports + heliports across the USA

Since pilots will be deployed across the USA, use the national file. It was processed from the FAA Aeronautical Data Delivery Service: filtered to operational facilities only, Z-coordinates stripped, properties normalized, and a tier field

added to every feature for zoom-level visibility control.

★

USE THIS FILE FOR ALL USA DEPLOYMENTS

rc-airports-national.geojson

10,642 features | 2.89 MB | Airports (AD) + Heliports (HP)

Three scoped files are available — use the national file for production. The state/region files are useful for local testing only.

FILE	FEATURES	SIZE	USE
rc-airports-national.geojson ★	10,642	2.89 MB	PRODUCTION — all USA pilots
rc-airports-southeast.geojson	2,495	660 KB	Testing — SE region only
rc-airports-georgia.geojson	235	62 KB	Testing — Georgia only

Upload to Mapbox Studio:

Studio → Tilesets → New tileset → Upload GeoJSON → Name it rc-airports → Wait for processing (~60-90 sec at 2.89 MB)

Understanding the Tier Field

Every feature in the national file includes a tier property (1-4) computed from FAA attributes. Tier controls which airports appear at each zoom level — essential for keeping the national view clean when dispatchers are looking at a broad operational area.

TIER	DEFINITION	COUNT	FIRST VISIBLE AT
1	Major — ICAO code + instrument approach (ATL, ORD, LAX, DFW, MIA, etc.)	2,268	Zoom 4
2	Regional — instrument approach, no ICAO	783	Zoom 7
3	Public GA — public airport, no IAP	1,935	Zoom 9
4	All heliports incl. hospital pads + private airports	5,656	Zoom 10

Circle layer configuration:

```

Layer ID: rc-airports-circle
Type: Circle
Source: rc-airports

# Color by type: amber=airport, blue=heliport
circle-color: match type: AD=#f59e0b HP=#0ea5e9
circle-radius: match type: AD=5 HP=4
circle-stroke-width: 1.5
circle-stroke-color: #0c0b14

# Zoom-based opacity using tier – keeps national view clean
circle-opacity:
  ["step", ["zoom"],
    0, zoom <4: nothing visible
    4, match tier: 1 -> 0.85, 0, zoom 4-6: Tier 1 only (major)
    7, match tier: [1,2] -> 0.85, 0, zoom 7-8: + regional
    9, match tier: [1,2,3] -> 0.85, 0, zoom 9: + public GA
    10, 0.85] zoom 10+: all incl. heliports

```

Label layer configuration (start one zoom level after circles):

```

Layer ID: rc-airports-label
Type: Symbol
Source: rc-airports

text-field: {ident} (FAA identifier – ATL, ORD, FTY, etc.)
text-size: 10
text-offset: [0, 1.6]
text-anchor: top
text-color: #f59e0b
text-halo-color: #0c0b14
text-halo-width: 1.5

# Same zoom/tier logic for text – one zoom level later than circles
text-opacity: ["step", ["zoom"], 0, 5, match tier:1->0.9,0,
  8, match tier:[1,2]->0.9,0, 10, match tier:[1,2,3]->0.9,0, 11, 0.9]

```

Heliport (HP) features include hospital rooftop and ground pads — e.g. Children's Healthcare of Atlanta, St. Mary's, Grady Memorial. These are Tier 4 and appear only at zoom 10+ to avoid clutter at regional view, but are critical for EMS aeromedical routing when dispatchers zoom into an incident area.

4 Agency / PSAP Zones

Draw your service area and upload as a GeoJSON tileset

This is the most operationally important layer for 911 deployments. Each agency needs a polygon representing their PSAP coverage boundary.

Option A — Draw in geojson.io (fastest for dev/demo):

1. Go to <https://geojson.io>
2. Use the polygon draw tool to trace the agency's service boundary
3. Click the feature, add a property: "name": "Agency Name"
4. Save → GeoJSON → upload to Mapbox as tileset named rc-agency-zones

Option B — Convert from shapefile (existing agency GIS data):

Drag the shapefile into <https://mapshaper.org>, export as GeoJSON, then upload to Mapbox.

Layer configuration:

```
# Fill layer
Layer ID:    rc-agency-zones-fill
Type:        Fill
fill-color:   #1a2a4a
fill-opacity: 0.18

# Line layer
Layer ID:    rc-agency-zones-line
Type:        Line
line-color:   #3b82f6
line-width:   1.5
line-opacity: 0.7
```

5 Venue & Campus Zones

Add per client during onboarding — not in the dev style

Venue and campus zone polygons are customer-specific and should be added to customer-specific Mapbox styles, not the shared dev style. During onboarding, trace the facility boundary and any sub-zones (gates, concourses, parking areas, buildings) using geojson.io.

GeoJSON feature shape:

```
{
  "type": "Feature",
  "properties": {
    "name": "Gate C",
    "zone_type": "entrance", // outer | gate | concourse | parking
    "section": "C"
  }
}
```

Layer configuration:

```
# Venue
rc-venue-zones-fill    fill-color: #3a1a06  fill-opacity: 0.20
rc-venue-zones-line    line-color: #f59e0b  line-width: 1.5

# Campus
rc-campus-zones-fill    fill-color: #0a1810  fill-opacity: 0.20
rc-campus-zones-line    line-color: #10b981  line-width: 1.5
```

6 Layer Order in Studio

Render order controls what appears on top

Mapbox renders layers bottom-up. The Studio layer panel list order determines what's on top. Configure in this order (top = renders on top):

```
rc-airports-label      ← top (labels always readable)
rc-airports-circle
rc-live-traffic-closures ← closures render above traffic flow
rc-live-traffic
rc-state-boundaries-line
rc-counties-line
rc-venue-zones-line
rc-venue-zones-fill
rc-campus-zones-line
rc-campus-zones-fill
rc-agency-zones-line
rc-agency-zones-fill    ← bottom (fills render under lines)
[base map layers]

# Live incidents added by app at runtime – always render on top
```

7 Publish & Verify

Final checks before committing to production

Publish the style

Click Publish in the top-right of Mapbox Studio. Confirm the Style URL has not changed:
mapbox://styles/memphman2004/cmr3afd69002401qq1uywfk5p

Add style URL to env

Add NEXT_PUBLIC_MAPBOX_STYLE_URL to scripts/env-web-ssr-prod.sh alongside the existing token variable.

Run the app locally

Start the dev server and open any dashboard page with the RapidCortexMap component.

Check dev console

Look for '[RapidCortexMap] Layer X not in loaded style' warnings. One warning per missing layer is expected until all tilesets are uploaded. No warnings = all layers confirmed.

Test layer toggles

Open the Map Layers panel and toggle each group. Confirm the layers appear/disappear correctly on the map.

Test incident markers

Mock incidents from map-utils.ts should appear as colored circles. Click one — confirm the popup renders with correct data.

ENVIRONMENT VARIABLES

```
# .env.local (development)
NEXT_PUBLIC_MAPBOX_ACCESS_TOKEN=pk.eyJ1Ijo... # public token
NEXT_PUBLIC_MAPBOX_STYLE_URL=mapbox://styles/memphman2004/cmr3afd69002401qq1uywfk5p

# scripts/env-web-ssr-prod.sh (add alongside existing token block)
export NEXT_PUBLIC_MAPBOX_STYLE_URL="mapbox://styles/memphman2004/cmr3afd69002401qq1uywfk5p"
```

QUICK REFERENCE

Minimum Viable Map (get it working today)

If you want the map functional with geographic context for a demo or pilot without completing all layers:

TASK	TIME	PRIORITY
Counties + state layers (built-in source)	10 min	DO FIRST
Agency zone (draw on geojson.io)	15 min	DO FIRST
Add NEXT_PUBLIC_MAPBOX_STYLE_URL to env	2 min	DO FIRST
Airports (upload rc-airports-national.geojson)	10 min	DO NEXT
Venue/campus zones	Per client	PER ONBOARDING

External Resources

Mapbox Studio: <https://studio.mapbox.com>

Mapbox token manager: <https://account.mapbox.com/access-tokens>

Draw GeoJSON: <https://geojson.io>

Convert shapefiles: <https://mapshaper.org>

FAA airport data: https://adds-faa.opendata.arcgis.com/datasets/e747ab91a11045e8b3f8a3efd093d3b5_0/explore

US Census boundaries: <https://www.census.gov/geographies/mapping-files/time-series/geo/cartographic-boundary.html>